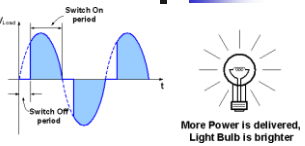
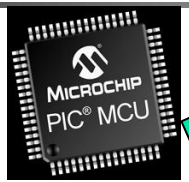
Part II ..... Agenda

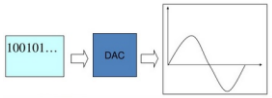


Load Power Control



Communication

Pulse Width Modulation (PWM)



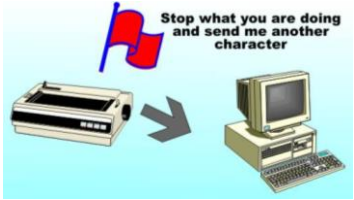
Digital to Analog Converter (DAC)

Part II

Analog to Digital Conversion (ADC)

Analog (input) Sensor

Interrupts



MCU: please take care !!!

Timers

Precise Timing Operations

Course Part II ILO's

Upon successful completion of this part of the course, the student will be able to:

- **Implement a UART Communication between MCUs and PC.**
- **Read and Process an analog signals using MCU.**
- **Use timers to produce a precise timing.**
- **Recognize interrupts and how to implement inside MCU.**
- **Generate PWM signal to control the power delivered to a load.**
- **Work cooperatively on teams.**
- Prepare and present **technical reports.**

Course Project

Course Assessment

- **Quiz(zes) on part II**

	Instructor "1"	Instructor "2"	Assistant
Midterm	10		
Quiz		10	
Lab Work			10
Practical Exam	10		
Final Exam	60		

- **10 Marks: Project on both Parts I & II** (*It will be delivered to & evaluated by the **Teaching Assistant***)

Course Project

Topics

- **Implementation of a mini embedded system that satisfies the following criteria:**
  - Solves a mini real world problem.
  - Utilizes the capabilities of MCU such as reading and processing of digital and analog inputs, communication with PC for monitoring and remote control, control a digital output, Interrupts, Timers, and PWM.
- **Some suggested topics [Optional]**
  - Control direction of a DC motor via hand gesture + Monitoring.
  - Control direction of a DC motor via head gesture + Monitoring.
  - Temperature ON/OFF control of a box (incubator) + Monitoring.
  - Human Detection using Sonar sensor + Motor (on/off) + buzzer + Monitoring.
  - Water Level Indicator.
  - Street Light that Glows on Detecting Vehicle Movement.

Course Project

Teams

- **The team should include 10 students who will divide themselves into sub-teams working on the software and hardware developments.**

Project Presentation & Evaluation:

- **Each team must prepare a CD for the course Teaching Assistant containing all materials related to the project (Well-documented project technical report + Software). *Only printed copy of technical report*.**
- **The project evaluation will be based on the following criteria:**
  - **Project management [10%]:** Cooperation – Duties distribution – Problem solving – Team players
  - **Software [40%]:** Open source or your own - Ease of use and operability - Accuracy - Output graphics.
  - **Experimental Hardware [30%]**: Stability and reusability – Precision - Cost optimization.
  - **Final Report [20%]:** Technical report format [Cover, TOC (Table of Contents), Introduction, Motivation, Objectives, Methodology, Results (Simulation + Experimental), References].
  - **Novelty [10% bonus]:** You have to prove to what extend the idea is novel.

What is expected of you in this course?

- **Download all handouts from the specified internet site and check it frequently.**
- **Attend all Lectures and Labs.**
- **Read the assigned portions of the book (The lectures are not a substitute for reading the book).**
- **Do the assignments and submit them.**
- **Assignment(s) on Part II [Optional – No Grading] (***They will be delivered (checked) to (by) Teaching Assistant***)**
- **Do the project and submit it in a standard form.**
- **You really must start the projects as soon as possible - you cannot throw them together at the last minute and have a good result!**